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1. Find a formula for the general term a_n of the sequence assuming that the pattern of the first few terms continues

$$\{4, -1, \frac{1}{4}, -\frac{1}{16}, \frac{1}{64}, \dots\}$$

2. Determine whether the sequence converges or diverges. If it converges, find the limit:

a) $a_n = 2 + (\frac{1}{\pi})^n$,

b) $a_n = \frac{3\sqrt{n}}{\sqrt{n+2}}$,

c) $a_n = \cos(\frac{n\pi}{n+1})$,

d) $a_n = \frac{(2n-1)!}{(2n+1)!}$,

e) $a_n = n - \sqrt{n+1}\sqrt{n+3}$,

f) $a_n = (1 - e^{-\frac{1}{\sqrt{n}}}) \sin(\frac{1}{n})$,

g) $a_n = \frac{n!}{2^n}$, **Hint:** For all $m \geq 2$, $\frac{m}{2} \geq 1$.

h) $a_n = (-1)^n$.

3. Determine whether the sequence is increasing, decreasing or not monotonic. Is the sequence bounded?

$$a_n = \frac{1-n}{2+n}$$

4. A sequence (a_n) is given by

$$a_1 = 2, \quad a_{n+1} = \frac{1}{3 - a_n}$$

. Show that $0 < a_n \leq 2$ and it is decreasing. Deduce that the sequence is convergent and find its limit.

begin enumerate

5. Find a formula for the n partial sum of the following series and use it to find the series' sum if the series converges:

$$\frac{4}{1 \cdot 2} + \frac{5}{2 \cdot 3} + \frac{5}{3 \cdot 4} + \dots + \frac{5}{n(n+1)} + \dots$$

6. Determine whether the series is convergent or divergent. If it is convergent, find its sum.

a) $\sum_{n=1}^{\infty} \frac{n^2}{n^2 - 2n + 5}$.

b) $\sum_{n=1}^{\infty} \frac{2^n + 4^n}{e^n}$

7. If the n th partial sum of a series $\sum_{n=1}^{\infty} a_n$ is

$$s_n = \frac{n-1}{n+1}$$

Find a_n and $\sum_{n=1}^{\infty} a_n$.

8. Find the value of c if $\sum_{n=2}^{\infty} (1+c)^{-n} = 2$
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9. Does the following series converge or diverge?

$$\sum_{n=1}^{\infty} n^2 e^{-n^2}$$

10. Find the sum of the series $\sum_{n=1}^{\infty} n e^{-2n}$ correct to four decimal places.
11. Determine if the following series are convergent or divergent:

a) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}-1}$

b) $\sum_{n=1}^{\infty} \frac{2}{\sqrt{n}+2}$

c) $\sum_{n=1}^{\infty} \frac{n \sin^2 n}{1+n^3}$